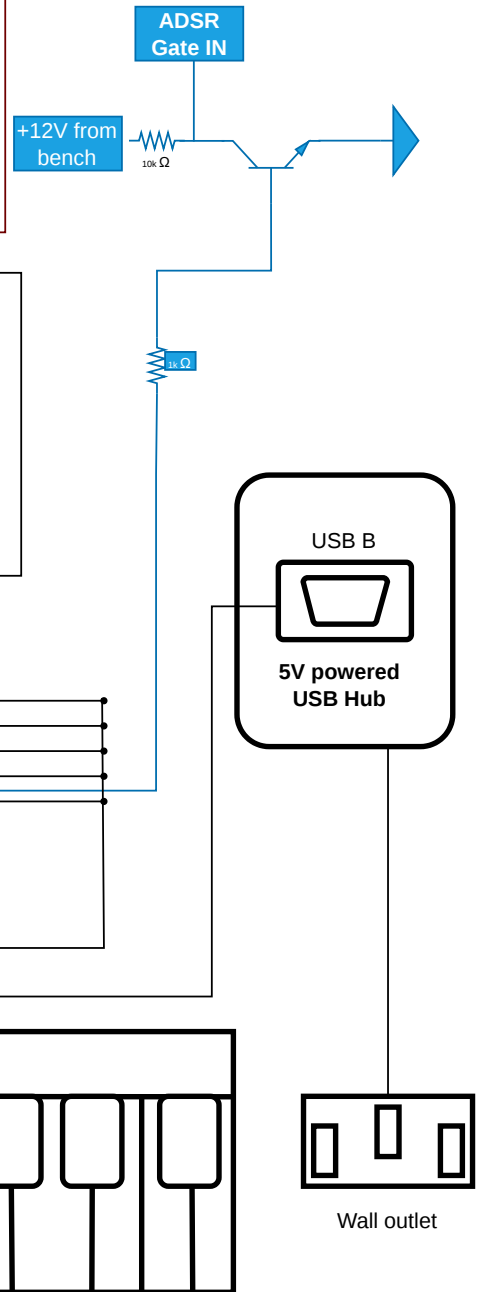


TL072
1IN+

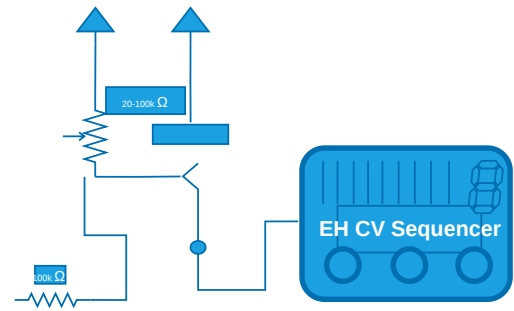
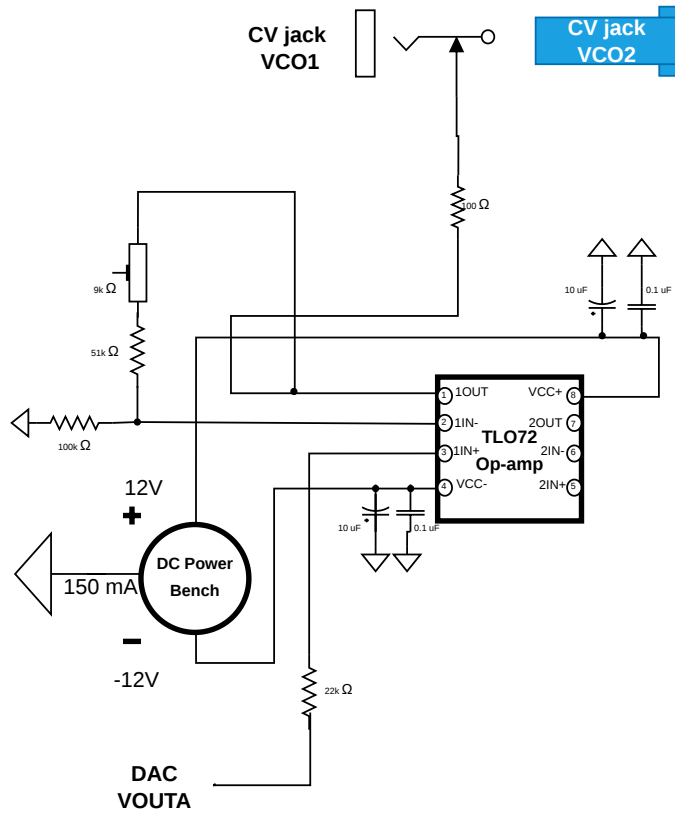
Teensy/DAC stage

BOM	
1	Alesis v25 MIDI Keyboard
1	Amazon Basics 7 port USB 5V 4A Hub
1	Teensy 4.1 Microcontroller
1	Teensy USB Host Cable Female
1	EXH 8-step program CV sequencer
1	AD5684BRUZ 12-Bit quad channel digital to analog converter
2	10k resistors
4	0.1 μ F ceramic capacitors
2	10 μ F electrolytic capacitor
1	LT1019ACN-2.5 VREF low noise 2.5V



***NOTE: ALL grounds shown MUST be tied to the same ground on the breadboard**

Op amp 1st stage



BOM	
1	TLO72IC dual rail to rail op amp
1	10k Ω trim pot (set at 9k)
1	T
1	T
1	Jesverty DC Power supply 30V 10A
1	A
1	T
4	Resistors: (1)22k Ω (1) 100 Ω (1) 100k Ω , (1) 51k Ω
4	0.1 μ F ceramic capacitors
2	10 μ F electrolytic capacitors